CLUSTERING OF COUNTRIES BASED ON SOCIO-ECONOMIC INDICATORS

Machine Learning Project

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PROBLEM STATEMENT

- GIVE Foundation is an NGO that provides financial aid to underdeveloped countries.
- After receiving $10 million in funding, the CEO must decide which countries need aid the most.
- Goal: Use machine learning clustering techniques to categorize countries and identify the countries that require the most financial assistance.

DATASET OVERVIEW

- Total Countries: 167
- Total Variables: 10

Important Variables:

- gdpp – GDP per capita
- income – Net income per person
- child_mort – Child mortality rate
- life_expec – Life expectancy
- inflation – Inflation rate

EXPLORATORY DATA ANALYSIS

EDA helps understand data distribution and detect patterns.

Key observations:

- GDP and Income distributions are right-skewed
- Few countries have very high economic indicators
- Majority of countries have moderate or low economic values

This reflects global economic inequality among countries.

FEATURE SCALING

- Feature scaling was applied using StandardScaler.
- Reason: Variables have different ranges (GDP up to 100000, child mortality below 200).
- Scaling ensures that all features contribute equally to clustering.

FINDING OPTIMAL CLUSTERS

- Elbow Method was used to determine the optimal number of clusters.
- WCSS was calculated for multiple values of K.
- The elbow point indicates the optimal number of clusters.
- Result: Optimal clusters = 3.

K-MEANS CLUSTERING

- After determining K=3, K-Means clustering was applied.
- Cluster interpretation:
- Cluster 0 – Developing countries
- Cluster 1 – Developed countries
- Cluster 2 – Underdeveloped countries
- Cluster 2 countries require the most financial aid.

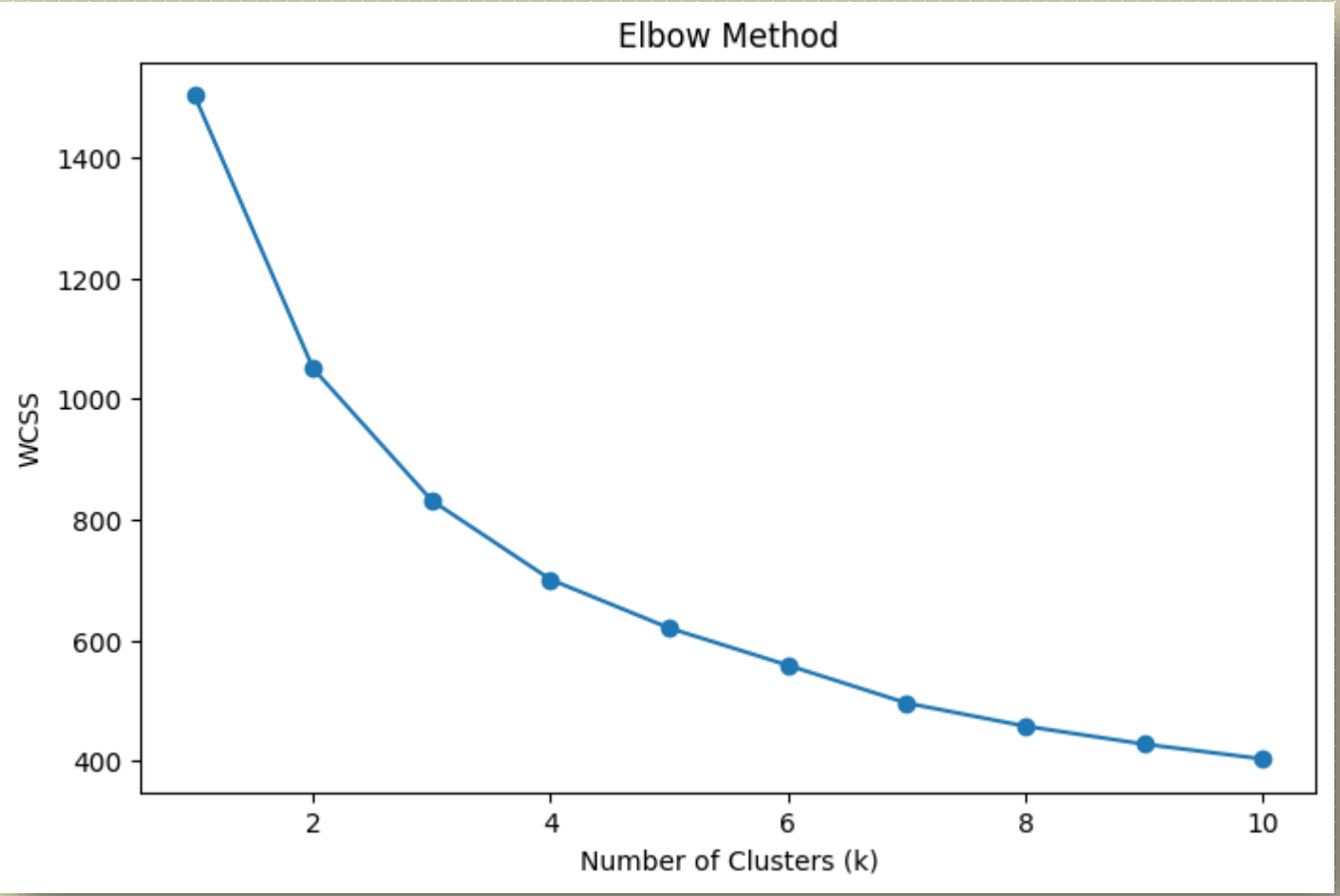
CLUSTER VISUALIZATION

- Scatter plots were used to visualize clusters:
- - GDP vs Child Mortality
- - Income vs GDP
- - Income vs Child Mortality
- These visualizations show clear separation between country groups.

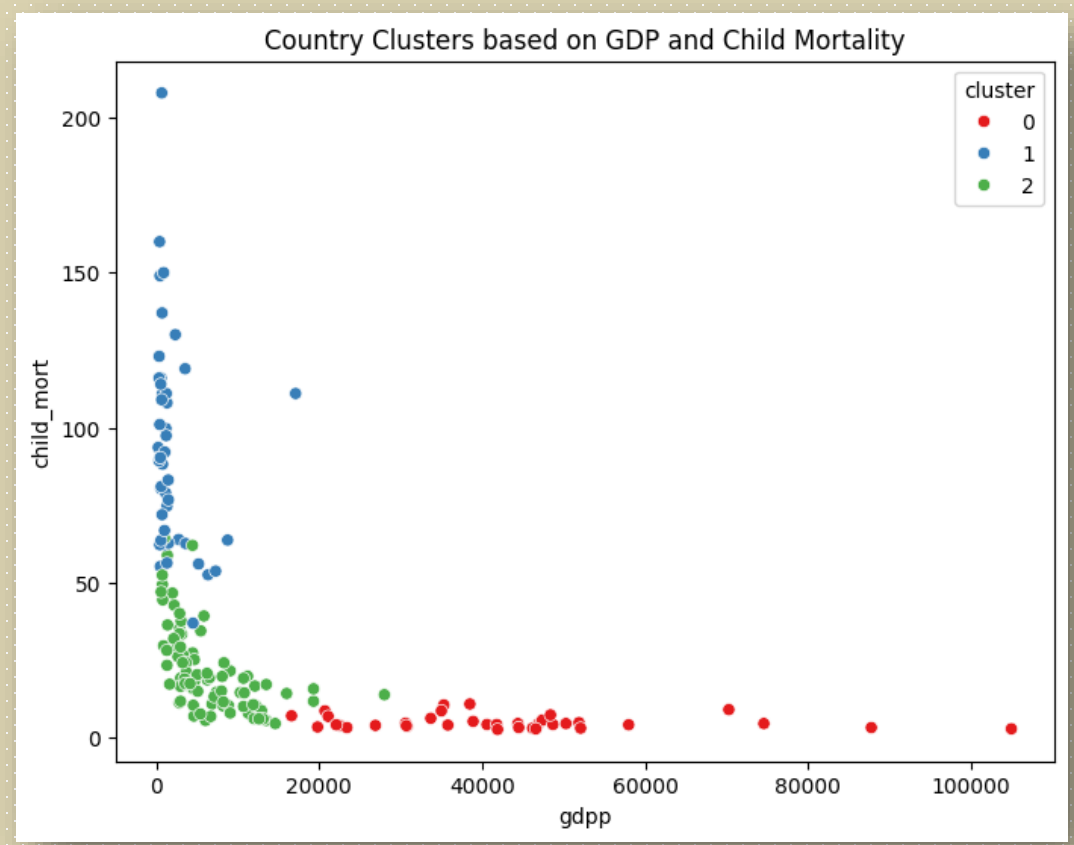
COUNTRIES RECOMMENDED FOR FINANCIAL AID

- Burundi
- Liberia
- Congo (Democratic Republic)
- Niger
- Sierra Leone

FINDING OPTIMAL CLUSTERS

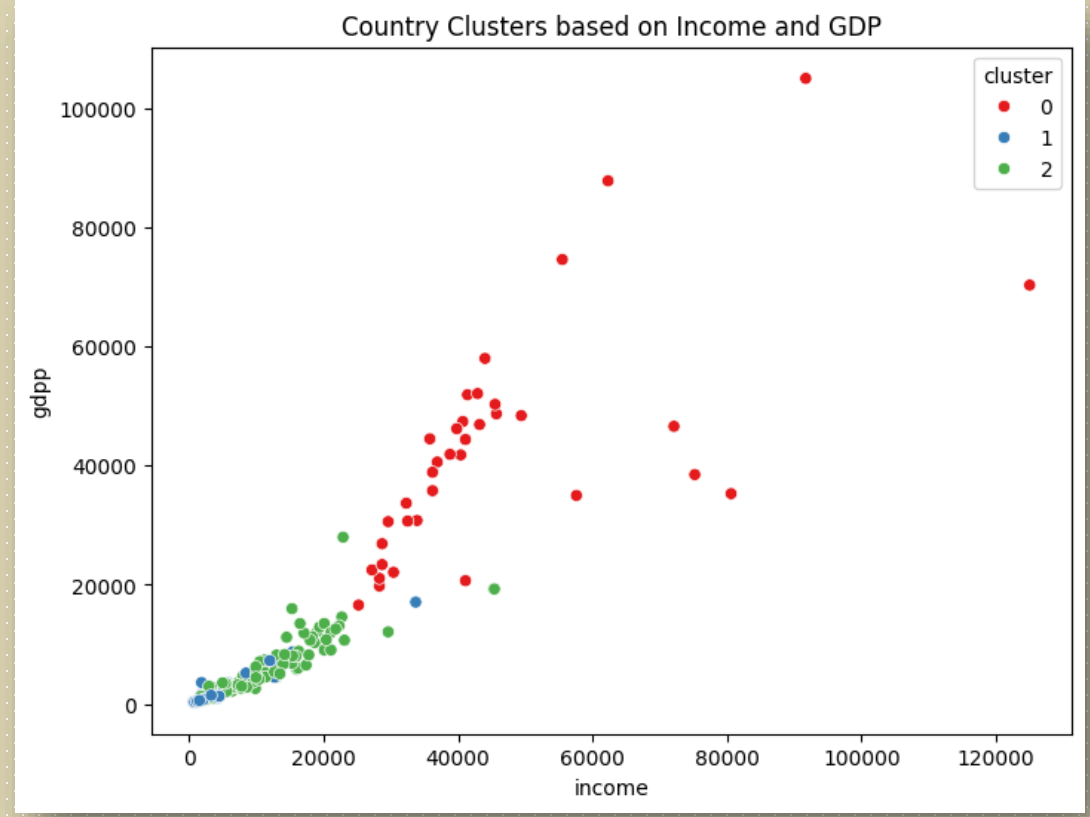


CLUSTER VISUALIZATION: GDP VS CHILD MORTALITY



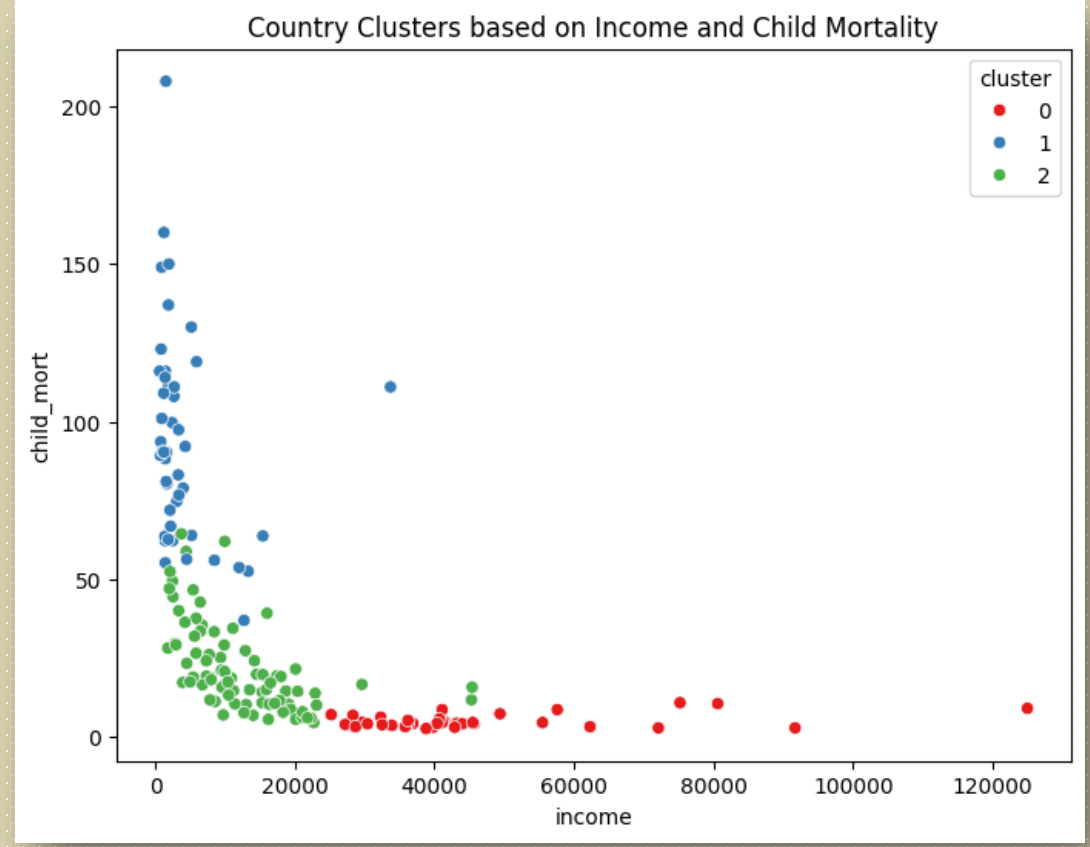
Observation: Countries with lower GDP per capita tend to have higher child mortality rates.

CLUSTER VISUALIZATION: INCOME VS GDP



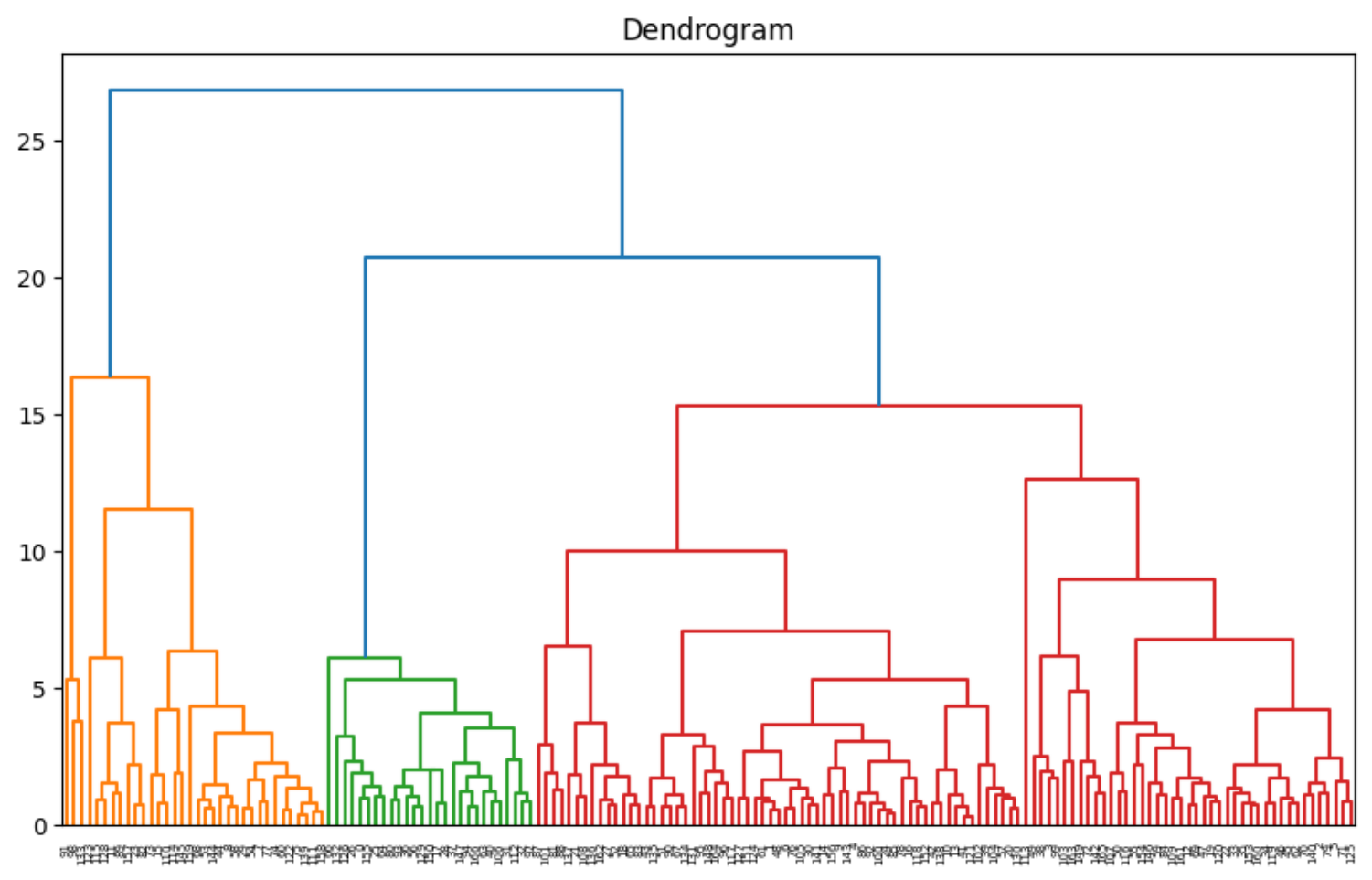
Observation: Countries with higher income generally have higher GDP per capita.

CLUSTER VISUALIZATION: INCOME VS CHILD MORTALITY



Observation: Lower income countries tend to show higher child mortality rates.

HIERARCHICAL CLUSTERING: DENDROGRAM



CONCLUSION

- Countries were successfully categorized into three development groups using clustering techniques.
- Cluster 2 countries have very low income and GDP levels along with high child mortality rates.
- These countries should be prioritized for financial aid by the NGO.
- Machine learning clustering helped identify the countries that require the most financial assistance.